

Controlled polynomial CFAR and linear Pauli–Fierz forest reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

This note advances the R-082 frontier in two directions while retaining its honesty boundary.

For FAR, the root-preserving six-coordinate current is used rather than the state-dependent four-row compression. The three polynomial currents $J_A = Dm_A$ are shown to vanish identically in the complete controlled stopped FAR wedge, including the moving endpoint, the low endpoint, and both predictable control drifts. The remaining controlled CFAR object is written as one exact input-scale telescope of three Cartan currents. This is a real closed subbranch and an exact reduction, not a bound for the remaining Cartan telescope. The canonical K_k smoothing closes the control-coordinate square function once, but a production scalar-ray fixture proves that it does not create the output orthogonality assumed in the sufficient R-082 Carleson lemma.

For NEAR, the three linear rows of the R-082 Pauli–Fierz coordinate are separated exactly from the rational row. Their Gram coefficient, target-heat lift, backward compensator, quadratic secant, conditional cross-covariance, and complete nine-block Wiener forest are now explicit. This finishes their algebraic bookkeeping. It does not make them positive under adapted future feedback. A heat-lifted Gaussian fixture leaves the rational row identically zero and produces a strictly negative zero chaos in one horizontal linear row. Thus the rational row is not the sole signed NEAR obstruction.

All statements are finite-cutoff statements for the hash-pinned A1 production coefficients, fixed positive density floor, common real-even regulators, exact non-aliased products, and the R-079 regular strict-past one-shot control class where a control statement is made. Controlled Cartan CFAR, a signed linear- or rational-row NEAR lower bound, the complete regular packet lower bound, overlap-stable progression, controlled-shell one-use, $q = 10/9$ Nelson, and Sector-A closure remain open.

2 Authorities and fixed coordinates

The load-bearing authorities are R-050 for the enhanced production current, R-063 for the complete lower-chaos forest, R-069–R-070 for the raw/Wick current telescope, R-075 for the production metric, R-079 for the exact full safe packet, R-080 for FAR square completion, R-081 for the polynomial–Cartan split, and R-082 for the stopped-current representation and global Pauli–Fierz coordinate.

The A1 manifest gives

$$P = M_X^2 + \epsilon_\rho = 4 + 10^{-12},$$
$$Q_{\Pi} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a = \frac{9}{500P}, \quad b = \frac{3}{400P}, \quad c = \frac{3}{320P}. \quad (2.1)$$

Diagonalisation is derived, rather than inserted:

$$\alpha = \frac{c}{b+c} = \frac{5}{9}, \quad c_1 = \frac{c}{a^2} = \frac{243}{8000P}, \quad c_0 = a - \frac{b^2}{c} = \frac{3}{250P}. \quad (2.2)$$

For each embedded Pauli generator put

$$m_A(z) = z^T S_A z, \quad F_A(z) = q_A(z)z, \quad q_A(z) = \frac{m_A(z)}{|z|^2 + \epsilon_\rho}. \quad (2.3)$$

For a translation H of the terminal Gaussian U , define

$$\mathcal{P}_A(H) = Dm_A(U + H), \quad \mathcal{C}_A(H) = F_A(U + H)^T D(U + H), \quad (2.4)$$

and

$$\mathcal{L}_A(H) = \mathcal{P}_A(H) - 2\alpha\mathcal{C}_A(H). \quad (2.5)$$

The fixed root-preserving current is

$$\mathcal{I}_T(H) = (\sqrt{c_0}\mathcal{P}_A(H), \sqrt{c_1}\mathcal{L}_A(H))_{A=1}^3. \quad (2.6)$$

Unlike the four-row Pauli–Fierz compression, (2.6) is related to the R-079 $Q_{\text{II}}^{1/2}$ current by one fixed orthogonal transformation. It therefore preserves every conditional root block.

3 The complete controlled polynomial FAR subbranch closes

Let $\ell = j_0 - 1$ and

$$A^{(k)} = A^{(\ell)} + \sum_{r=j_0}^k a_r, \quad a_k = K_k h_k \in \mathcal{F}_{k-1}. \quad (3.1)$$

Use the sharp cubes

$$Q_j = \{p \in \mathbb{Z}^3 : |p|_\infty \leq 2^j\}, \quad (3.2)$$

and suppose $U_j, A^{(j)}$ are supported in Q_j . Let Π_m be supported where $|p|_\infty > 2^{m-1}$.

Theorem 3.1 (controlled polynomial stopped FAR is zero)

For every $C \geq C_{\text{poly}} = 3$, every stopped index

$$n_C(m) = \min\{J, m - C\},$$

and every generator A , the complete controlled stopped bracket of $\mathcal{P}_A = Dm_A$ obeys

$$\boxed{\Pi_m R_{A, n_C(m)}^{\mathcal{P}} = 0.} \quad (3.3)$$

This includes the moving endpoint, the low endpoint, the predictable control-value drift, and the predictable control-derivative drift.

Proof. The current $Dm_A(z) = 2(S_A z)^T D z$ is quadratic. If z is supported in Q_n , its current is supported in Q_{n+1} . Conditional expectation over future Gaussian shells preserves this support. The only possible future contraction is $\mathbb{E}[V^T S_A D V]$, which vanishes for a common real-even covariance; even without that cancellation it would be a zero mode. The same argument applies to each conditional input increment and to both predictable drift channels. For $m \geq n + 3$,

$$2^{m-1} \geq 2^{n+2} > 2^{n+1},$$

so Π_m kills every term. Terminal saturation $n_C(m) = J$ changes no support inequality. This proves (3.3). $\square$

The safety choice $C_{\text{poly}} = 3$ avoids any boundary convention at $|p|_\infty = 2^{n+1}$. The theorem strengthens the R-081 polynomial support observation to the entire R-082 controlled stopped object; no drift has been silently discarded.

4 Exact Cartan input-scale telescope

Define terminal input increments before probability projection by

$$\Delta_{A,\ell} = \mathcal{C}_A(A^{(\ell)}) - \mathcal{C}_A(0), \quad (4.1)$$

and, for $k \geq j_0$,

$$\Delta_{A,k} = \mathcal{C}_A(A^{(k)}) - \mathcal{C}_A(A^{(k-1)}). \quad (4.2)$$

They give the pathwise input telescope

$$\mathcal{C}_A(A^{(n)}) - \mathcal{C}_A(0) = \Delta_{A,\ell} + \sum_{k=j_0}^n \Delta_{A,k}. \quad (4.3)$$

Let $P_j = \mathbb{E}[\cdot | \mathcal{F}_j]$ and $d_j = P_j - P_{j-1}$.

Theorem 4.1 (nonduplicating stopped input telescope)

The complete controlled Cartan bracket is exactly

$$\begin{aligned} R_{A,n} &:= P_n \{ \mathcal{C}_A(A^{(n)}) - \mathcal{C}_A(0) \} - P_\ell \Delta_{A,\ell} - \sum_{k=j_0}^n P_{k-1} \Delta_{A,k} \\ &= (P_n - P_\ell) \Delta_{A,\ell} + \sum_{k=j_0}^n (P_n - P_{k-1}) \Delta_{A,k} \\ &= \sum_{j=j_0}^n d_j \left(\Delta_{A,\ell} + \sum_{k=j_0}^j \Delta_{A,k} \right). \end{aligned} \quad (4.4)$$

Both endpoints occur exactly once. Equation (4.4) is a finite algebraic identity, not optional stopping.

Proof. Apply P_n to (4.3), subtract $P_\ell \Delta_{A,\ell}$ and the predictable increments $P_{k-1} \Delta_{A,k}$. Alternatively expand $P_n - P_{k-1} = \sum_{j=k}^n d_j$ and interchange the finite sums. $\square$

For $Z_k^- = U + A^{(k-1)}$, each input has the exact value/derivative split

$$\begin{aligned} \Delta_{A,k}^{\text{val}} &= \{ F_A(Z_k^- + a_k) - F_A(Z_k^-) \}^T D Z_k^-, \\ \Delta_{A,k}^{\text{der}} &= F_A(Z_k^- + a_k)^T D a_k. \end{aligned} \quad (4.5)$$

Writing $z_k^- = U_{k-1} + A^{(k-1)}$ and $\Phi_{A,k-1} = P_{\Sigma_{k-1}} F_A$, common real-even covariance gives

$$\begin{aligned} P_{k-1} \Delta_{A,k}^{\text{val}} &= \{ \Phi_{A,k-1}(z_k^- + a_k) - \Phi_{A,k-1}(z_k^-) \}^T D z_k^-, \\ P_{k-1} \Delta_{A,k}^{\text{der}} &= \Phi_{A,k-1}(z_k^- + a_k)^T D a_k. \end{aligned} \quad (4.6)$$

These are exactly the two predictable drifts in R-082.

For every $k \leq j$, the root $d_j \Delta_{A,k}$ is the difference of the R-082 three-channel formula at backgrounds with and without a_k . With $g_j = U_j - U_{j-1}$ and $\Phi_j = P_{\Sigma_{j+1}} F_A$, write

$$\begin{aligned} \iota_{j,k}^\Delta &= [\Phi_j(Z + a_k + g_j) - \Phi_j(Z + g_j)]^T D g_j, \\ \nu_{j,k}^\Delta &= [\Phi_j(Z + a_k + g_j) - \Phi_j(Z + a_k)]^T (DZ + D a_k) \\ &\quad - [\Phi_j(Z + g_j) - \Phi_j(Z)]^T DZ, \\ \kappa_{j,k}^\Delta &= [\Phi_j - \Phi_{j-1}](Z + a_k)^T (DZ + D a_k) - [\Phi_j - \Phi_{j-1}](Z)^T DZ. \end{aligned} \quad (4.7)$$

Then

$$d_j \Delta_{A,k} = \iota_{j,k}^\Delta + \nu_{j,k}^\Delta + \kappa_{j,k}^\Delta, \quad (4.8)$$

$$\mathbb{E}_{j-1} \iota_{j,k}^\Delta = 0, \quad \mathbb{E}_{j-1} (\nu_{j,k}^\Delta + \kappa_{j,k}^\Delta) = 0. \quad (4.9)$$

Thus raw value innovation is never called centered without its heat compensator.

Combining (2.5), (3.3), and (4.4) proves the exact controlled-CFAR reduction

$$\boxed{S_C^{\text{ctrl}} = \frac{3}{80P} \sum_m \sum_{A=1}^3 \mathbb{E} \|\Pi_m R_{A,n_C(m)}\|_2^2, \quad C \geq 3.} \quad (4.10)$$

Indeed $4\alpha^2 c_1 = 3/(80P)$. Equation (4.10) closes all three polynomial rows and nothing more: the three Cartan rows on its right remain open.

5 The control-coordinate ledger and the orthogonality boundary

If

$$\|K_k\| \leq \kappa_K 2^{-2k}, \quad \|Da_k\|_2 \leq \kappa_B 2^k \|a_k\|_2, \quad (5.1)$$

then spatial shell orthogonality gives the genuine one-use coordinate bound

$$\sum_k \left\{ 2^{4k} \mathbb{E} \|a_k\|_2^2 + 2^{2k} \mathbb{E} \|Da_k\|_2^2 \right\} \leq \kappa_K^2 (1 + \kappa_B^2) \mathbb{E} \sum_k \|h_k\|_{\mathcal{H}_k}^2. \quad (5.2)$$

This is useful input control. It does not imply that nonlinear output increments $\Pi_m \Delta_{A,k}$ are pairwise orthogonal.

Proposition 5.1 (production output-orthogonality no-go)

On a scalar doublet ray at unit floor let

$$F(x) = \frac{x^3}{1+x^2}, \quad a_1 = \frac{1}{4} \cos x, \quad a_2 = \frac{1}{5} \cos 3x, \quad (5.3)$$

$$\Delta_1 = F(a_1), \quad \Delta_2 = F(a_1 + a_2) - F(a_1). \quad (5.4)$$

For the cosine coefficient

$$c_3(f) = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos 3x \, dx,$$

let $P_{|p|=3}$ denote the exact orthogonal projection onto the two harmonics $p = \pm 3$ (not the relative-FAR cutoff Π_3). Direct periodic quadrature gives

$$\begin{aligned} c_3(\Delta_1) &= 0.003619922102104775 \dots, \\ c_3(\Delta_2) &= 0.020785528647497703 \dots, \end{aligned} \quad (5.5)$$

and hence

$$\langle P_{|p|=3} \Delta_1, P_{|p|=3} \Delta_2 \rangle_{\text{mean}} = \frac{1}{2} c_3(\Delta_1) c_3(\Delta_2) = 0.00003762099727750446 \dots > 0. \quad (5.6)$$

This unit-floor computation lies exactly in the pinned production-floor family. If

$$F_\epsilon(x) = \frac{x^3}{\epsilon + x^2}, \quad F_\epsilon(\sqrt{\epsilon} y) = \sqrt{\epsilon} F(y), \quad (5.7)$$

then the physical inputs are $a_1 = \sqrt{\epsilon_\rho} \cos x/4$ and $a_2 = \sqrt{\epsilon_\rho} \cos 3x/5$. Each coefficient in (5.5) is multiplied by $\sqrt{\epsilon_\rho}$, so the positive mean cross becomes $3.762099727750446 \dots \times 10^{-17}$ at $\epsilon_\rho = 10^{-12}$. The structural failure is already exact for the leading production jet $F_0(x) = x^3$. For $a_1 = A \cos x$, $a_2 = B \cos 3x$,

$$c_3(\Delta_1) = \frac{A^3}{4}, \quad c_3(\Delta_2) = \frac{3}{2}A^2B + \frac{3}{4}B^3, \quad (5.8)$$

so the projected cross is positive for $A, B > 0$. Analytic continuity transfers this to the rational production coefficient for small amplitudes; (5.5) checks the full rational function directly.

Thus K_k smoothing and spatial input-shell orthogonality do not verify the output orthogonality hypothesis of R-082 Lemma 6.1. This fires the method no-go NG-2026-07-25-A13-K-SMOOTHING-OUTPUT-ORTHOGONALITY. The fixture contains no martingale projection, and its second input is not placed in the relative-FAR range. It therefore refutes only the global raw output pairwise-orthogonality premise as written. It is not a counterexample to (4.10), does not exclude a far-only or correlated martingale estimate, and does not refute a correlated/signed Cartan Carleson theorem or complete controlled CFAR.

6 The exact linear Pauli–Fierz subpacket

Use the pinned six-real ordering

$$z = (x_1, x_2, x_3, y_1, y_2, y_3) = (\operatorname{Re} \psi_1, \operatorname{Re} \psi_2, \operatorname{Re} \chi, \operatorname{Im} \psi_1, \operatorname{Im} \psi_2, \operatorname{Im} \chi), \quad (6.1)$$

and put $P_{\text{dbl}} = \operatorname{diag}(1, 1, 0, 1, 1, 0)$ and $c_S = c_0 + c_1$. Define

$$A_0 = 2\sqrt{c_0}P_{\text{dbl}}, \quad (6.2)$$

$$A_1 = 2\sqrt{c_S} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (6.3)$$

$$A_2 = 2\sqrt{c_S} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (6.4)$$

Then A_0 is symmetric, A_1, A_2 are skew, and

$$z^T A_0 \dot{z} = \sqrt{c_0} dr, \quad z^T A_1 \dot{z} = 2\sqrt{c_S} \operatorname{Re} h, \quad z^T A_2 \dot{z} = 2\sqrt{c_S} \operatorname{Im} h. \quad (6.5)$$

Let $C_L(z)$ have rows $z^T A_\nu$, $0 \leq \nu \leq 2$. Its Gram matrix is

$$B_L(z) = C_L(z)^T C_L(z) = \sum_{\nu=0}^2 A_\nu^T z z^T A_\nu. \quad (6.6)$$

The fourth row is

$$C_R(z) = 2\sqrt{c_1} g(z)^T, \quad g(z) = P_{\text{dbl}} z - \alpha \frac{z^T P_{\text{dbl}} z}{z^T z + \epsilon_\rho} z, \quad (6.7)$$

and the full R-082 Gram matrix splits exactly as

$$B(z) = B_L(z) + C_R(z)^T C_R(z). \quad (6.8)$$

For centered target heat with covariance Σ put

$$H_\Sigma = \sum_{\nu=0}^2 A_\nu^T \Sigma A_\nu. \quad (6.9)$$

Quadraticity gives

$$\boxed{P_\Sigma B_L(z) = B_L(z) + H_\Sigma}, \quad H_{\Sigma_1 + \Sigma_2} = H_{\Sigma_1} + H_{\Sigma_2}. \quad (6.10)$$

Thus the heat-lifted linear packet is

$$\mathcal{W}_{L,\Sigma}(z, y) = \frac{1}{2} \{B_L(z) + H_\Sigma\} : (y^{\otimes 2} - \Gamma). \quad (6.11)$$

The outer factor $1/2$ is load-bearing. Exposing a value shell transfers its $H_{\Delta\Sigma}$ exactly into B_L ; exposing a derivative shell changes Γ . The coefficient secant also closes exactly:

$$B_L(z + a) - B_L(z) = DB_L(z)[a] + B_L(a). \quad (6.12)$$

There is no higher coefficient Taylor remainder in the linear rows.

7 Complete forest and conditional covariance defects

For symmetric matrices S, T define the fixed Gram contraction

$$\mathcal{T}(S, T) = \sum_{\nu=0}^2 \text{Tr}(A_\nu^T S A_\nu T). \quad (7.1)$$

Let X, Y be centered jointly Gaussian exposed value and derivative vectors with $\text{Cov}(X, Y) = 0$, value covariance Σ_X , derivative covariance Γ , predictable controls A, b , and independent target-heat covariance Σ_h . Put

$$\begin{aligned} R_2 &= :XX:, & R_1 &= XA^T + AX^T, & R_0 &= AA^T + \Sigma_X + \Sigma_h, \\ C_2 &= :YY:, & C_1 &= Yb^T + bY^T, & C_0 &= bb^T. \end{aligned} \quad (7.2)$$

Theorem 7.1 (complete linear-row forest)

The heat-lifted covariance-normal packet is the exact nine-block identity

$$\boxed{2\mathcal{W}_L = \mathcal{T}(R_2 + R_1 + R_0, C_2 + C_1 + C_0)}. \quad (7.3)$$

The homogeneous block counts are

$$P_4 : 1, \quad P_3 : 2, \quad P_2 : 3, \quad P_1 : 2, \quad P_0 : 1. \quad (7.4)$$

Target heat supplies exactly

$$\frac{1}{2} \mathcal{T}(\Sigma_h, C_2), \quad \frac{1}{2} \mathcal{T}(\Sigma_h, C_1), \quad \frac{1}{2} \mathcal{T}(\Sigma_h, C_0). \quad (7.5)$$

These are block counts inside the linear coordinate. They do not replace the R-063 tensor-family classification for the complete production frame.

Proof. The sums in (7.2) are

$$R_2 + R_1 + R_0 = (X + A)(X + A)^T + \Sigma_h,$$

and

$$C_2 + C_1 + C_0 = (Y + b)(Y + b)^T - \Gamma.$$

Bilinearity of $\mathcal{T}$ gives all nine blocks and (7.3). $\square$

There are two distinct adapted covariance defects. For a terminal bilinear row $Z^T A_\nu Y$, let

$$Z_j = P_j Z, \quad Y_j = P_j Y, \quad K_j = P_j[(Z - Z_j)(Y - Y_j)^T]. \quad (7.6)$$

Then

$$P_j[Z^T A_\nu Y] = Z_j^T A_\nu Y_j + A_\nu : K_j, \quad (7.7)$$

and hence

$$\begin{aligned} d_j[Z^T A_\nu Y] &= Z_{j-1}^T A_\nu d_j Y + (d_j Z)^T A_\nu Y_{j-1} \\ &\quad + (d_j Z)^T A_\nu d_j Y + A_\nu : (K_j - K_{j-1}). \end{aligned} \quad (7.8)$$

The last line is a conditional value-derivative cross-covariance defect.

Even if $K_j = 0$, derivative covariance may remain signed. If $\mu = \mathbb{E}[Y \mid \mathcal{A}]$ and $V = \text{Cov}(Y \mid \mathcal{A})$, the difference of two linear-row packets has conditional expectation

$$\begin{aligned} \mathbb{E}[\Delta \mathcal{W}_L \mid \mathcal{A}] &= \frac{1}{2} \bar{B}_1 : \{V + (\mu + b)^{\otimes 2} - \Gamma\} \\ &\quad - \frac{1}{2} \bar{B}_0 : \{V + \mu^{\otimes 2} - \Gamma\}, \end{aligned} \quad (7.9)$$

where $\bar{B}_k = B_L(z_k) + H_\Sigma$. Under $\mu = 0$ and $V = \Gamma$, (7.9) reduces to the positive square $\bar{B}_1 : b^{\otimes 2}/2$. These are clean sufficient conditions (and the generic uniform ones); adapted future feedback does not guarantee them, though special configurations may cancel accidentally.

8 An adapted linear row has negative zero chaos

Let $\xi \sim N(0, 1)$, $\lambda \neq 0$, and

$$A = \lambda(\xi^2 - 4), \quad z = A e_1, \quad y = \xi e_2. \quad (8.1)$$

Allow an independent centered target-heat dummy $R = \tau \zeta e_1$. Then

$$dr = d\rho = 0, \quad h = (A + R)\xi. \quad (8.2)$$

The rational row, radial linear row, and imaginary horizontal row vanish identically. The entire packet is the real horizontal linear row

$$\mathcal{W} = 2c_S(A^2 + \tau^2)(\xi^2 - 1). \quad (8.3)$$

Writing H_k for probabilists' Hermites,

$$A^2 = \lambda^2(H_4 - 2H_2 + 11), \quad (8.4)$$

and the complete lower-chaos identity is

$$A^2 H_2 = \lambda^2(H_6 + 6H_4 + 15H_2 - 4). \quad (8.5)$$

Therefore

$$\mathcal{W} = 2c_S\lambda^2 H_6 + 12c_S\lambda^2 H_4 + (30c_S\lambda^2 + 2c_S\tau^2)H_2 - 8c_S\lambda^2, \quad (8.6)$$

and

$$\boxed{\mathbb{E}\mathcal{W} = -8c_S\lambda^2 < 0.} \quad (8.7)$$

Heat adds only the centered $2c_S\tau^2 H_2$ term. It cannot change the negative zero chaos.

A separate three-point law gives the same sign without Gaussian Hermite algebra:

$$\mathbb{P}(\xi = 0) = \frac{1}{2}, \quad \mathbb{P}(\xi = \sqrt{2}) = \mathbb{P}(\xi = -\sqrt{2}) = \frac{1}{4}, \quad (8.8)$$

and $A = 1 - \xi^2/2$ gives

$$\mathbb{E}[A^2(\xi^2 - 1)] = -\frac{1}{2}, \quad \mathbb{E}\mathcal{W} = -c_S. \quad (8.9)$$

This fires NG-2026-07-25-A13-LINEAR-PF-ADAPTED-POSITIVITY. It is an exact finite-root algebraic no-go to universal linear-row positivity and to declaring the rational row the sole signed obstruction. It is not a spatial Cameron–Martin counterexample and does not falsify a lower bound for the recombined, paid production packet.

9 Evidence map and successor

The accepted and failed routes now form the following map.

1. *Controlled polynomial FAR: closed exactly.* The fixed root-preserving coordinate and sharp-cube support eliminate all three J_A rows, including endpoints and drifts, for $C \geq 3$.
2. *Controlled Cartan FAR: exact target, open estimate.* Equation (4.4) assigns every input once and (4.10) is the complete remaining CFAR square. No inequality for that square is claimed.
3. *Control-coordinate smoothing: closed but insufficient.* Equation (5.2) spends Cameron–Martin input once. Equations (5.5)–(5.8) show that exact-harmonic nonlinear outputs are correlated, so R-082 global raw-output orthogonal Carleson cannot be invoked from this fact alone; a far-only correlated martingale estimate remains possible.
4. *Linear Pauli–Fierz algebra: closed.* Gram, heat, secant, cross-covariance, trace, and all nine forest blocks are explicit.
5. *Linear Pauli–Fierz adapted sign: failed as a standalone route.* The rational row can vanish while the linear packet has negative expectation. The linear and rational rows must remain together with the paid subtraction and complete spatial budget.

The next FAR attack must prove a correlated or signed paracomposition bound for the Cartan input telescope (4.4); imposing pairwise orthogonality is not available. The next NEAR attack must estimate both the linear covariance defects (7.8)–(7.9) and the rational secant–Jensen defect inside one heat-lifted square–trace–forest packet. Only after controlled CFAR and the complete signed NEAR estimate may the R-081 temporal Douglas factorisation be used for overlap-stable progression, followed by one-use and $q = 10/9$ Nelson.

10 Devil’s-advocate review

1. **Objection:** “R-081 already closed the complete polynomial controlled FAR bracket.” **VALID with mitigation.** R-081 proved the underlying support simplification. R-083 upgrades it to the exact R-082 stopped object and explicitly keeps the moving endpoint, low endpoint, and both drifts. No new analytic decay is attributed to the polynomial rows.
2. **Objection:** “Equation (4.4) duplicates early roots of a later control.” **DISMISSED.** The input k begins at $P_n - P_{k-1}$; roots earlier than k belong to the predictable drift and are subtracted once. The pre- j_0 input is separately $(P_n - P_\ell)\Delta_{A,\ell}$.
3. **Objection:** “The heat compensator was absorbed into raw value innovation.” **DISMISSED.** It is the explicit third line of (4.7), and only $\nu^\Delta + \kappa^\Delta$ is centered.
4. **Objection:** “The positive K_k ledger proves the R-082 Carleson hypothesis.” **UPHELD as false.** The exact cubic and full rational fixtures have positive exact-harmonic cross products, including after rescaling to the pinned production floor. They do not test a weaker far-only correlated martingale estimate.
5. **Objection:** “Three linear rows can be removed from NEAR because their coefficient is polynomial.” **UPHELD as false.** Polynomial heat algebra is finite, but adapted covariance defects are signed. The fixture (8.1) leaves only a linear horizontal row and has negative mean.
6. **Objection:** “The negative fixture disproves controlled-shell one-use.” **UPHELD as an overclaim.** It has no spatial shell or Cameron–Martin payment and therefore tests only universal rowwise sign.
7. **Objection:** “The nine blocks replace the R-063 forest.” **UPHELD as false.** They are the exact block expansion of the linear Gram tensor. The rational tensor families and the complete paid packet remain governed by R-063/R-079.

External adversarial review is invited for the support boundary $C = 3$, the low-input term in (4.4), the common-even contraction in (4.6), the factor $3/(80P)$, the two independent Fourier coefficient calculations, the outer $1/2$ in (6.11) and (7.3), and the zero-chaos sign in (8.6).

11 Reproduction and independent audit

The primary executable derives every production constant from the A1 manifest, verifies the six-real matrices and Gram split, performs an exact quadratic heat average, checks the secant and nine-block forest, executes a finite-tree conditional cross-covariance identity, derives the Gaussian Hermite coefficients, tests the discrete sign fixture, verifies the complete input telescope and three root channels, checks the K_k ledger, and computes the rational Fourier cross by a 2^{16} -point periodic rule.

The non-importing executable uses Decimal production arithmetic, exact Fraction filtrations and heat channels, a scalar forest derivation, normal moments and power matching instead of the primary Hermite conversion, the three-point law, exact cubic coefficients, and a separate 2^{15} -point periodic quadrature.

Run from the repository root:

```
python codes/foundations/  
a13_classii_controlled_polynomial_cfar_linear_pf_forest_  
reduction_verify.py
```

The child contracts are primary 58/58 and non-importing independent 43/43. The integrated verifier pins R-082 and the A1 manifest, compares independently computed constants and Fourier coefficients, checks every authority, note, PDF, manifest, result, negative-result, exploration, task, gate, roadmap, claim, lineage, and honesty surface, and fails closed on any promotion of the remaining gates.

Result footer

Result ID: A13-CLASSII-CONTROLLED-POLYNOMIAL-CFAR-LINEAR-PAULI-FIERZ-FOREST-REDUCTION

Precise statement: For $C \geq 3$, all three controlled polynomial J_A FAR rows vanish in the complete stopped-current object, with both endpoints and predictable drifts retained. Controlled CFAR reduces exactly to the three Cartan input-scale telescopes with coefficient $3/(80P)$. K-shell smoothing pays the input coordinate once but does not make nonlinear output increments orthogonal. The three linear Pauli--Fierz rows have an exact Gram, heat, secant, cross-covariance, trace, and nine-block forest algebra. An adapted heat-lifted fixture with zero rational row has negative linear-row zero chaos.

Scope: Finite cutoff; fixed positive floor; common real-even regulator; exact products; fixed root-preserving coordinate for CFAR; regular strict-past controls.

Dependencies: A1 manifest; R-050; R-063; R-069--R-070; R-075; R-079; R-080; R-081; R-082.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: python codes/foundations/a13_classii_controlled_polynomial_cfar_linear_pf_forest_reduction_verify.py

Expected output: Primary 58/58; independent 43/43; manifest-pinned integrated and aggregate PASS; exit zero.

Falsification gate: Failure of polynomial stopped support, either endpoint or drift, the Cartan input telescope or $3/(80P)$ factor, three-channel centering, rational cross fixture, linear Gram/heat/secant/forest, conditional defect, adapted sign, evidence hash, PDF contract, or honesty flag.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No controlled Cartan or complete CFAR estimate, signed linear/rational/complete NEAR lower bound, regular complete-packet lower bound, overlap-stable progressive extension, controlled-shell one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure is proved.

Next required action: Prove a correlated/signed Cartan input-scale CFAR estimate and a recombined linear+rational heat-lifted NEAR bound; then progression, one-use, and Nelson.